

Structure investigations on assembled astaxanthin molecules

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Abstract

The carotenoid *r,r*-astaxanthin (3*R*,3′*R*-dihydroxy-4,4′-diketo-β-carotene) forms different types of aggregates in acetone–water mixtures. H-type aggregates were found in mixtures with a high part of water (e.g. 1:9 acetone–water mixture) whereas two different types of J-aggregates were identified in mixtures with a lower part of water (3:7 acetone–water mixture). These aggregates were characterized by recording UV/vis-absorption spectra, CD-spectra and fluorescence emissions. The sizes of the molecular assemblies were determined by dynamic light scattering experiments. The hydrodynamic diameter of the assemblies amounts 40 nm in 1:9 acetone–water mixtures and exceeds up to 1 μm in 3:7 acetone–water mixtures. Scanning tunneling microscopy monitored astaxanthin aggregates on graphite surfaces. The structure of the H-aggregate was obtained by molecular modeling calculations. The structure was confirmed by calculating the electronic absorption spectrum and the CD-spectrum where the molecular modeling structure was used as input.

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1. Introduction

Carotenoids are largely abundant throughout flora, fauna, and micro-organisms [1]. The most noticeable property of carotenoids is the coloring of living beings. Astaxanthin (Fig. 1) is a red pigment which naturally occurs in most of the crustaceans and it is often used for the coloration of fish, especially salmon. Since carotenoids in general are relatively unstable molecules, they are commonly associated with proteins. An eye-catching example for the presence of carotenoid–protein complexes and the following reformation to monomer species is the change in color when lobsters are boiled. Grey-blue when alive, the lobster becomes deep red after cooking by setting astaxanthin free from the carotenoprotein complex [2].

The change in color as a result of specific aggregation has found a large economic importance for the coloring of

nutrition. This is performed by preparing artificial assemblies in which carotenoids are for instance embedded in gelatine nanoparticles [3]. Changes of color do also occur in systems in which carotenoids are simply self-assembled. Aggregation has been investigated for β-carotene [4], violaxanthin [5,6], astaxanthin [7,8], zeaxanthin [6,9], as well as for xanthophylls [10,11]. The formation of self-assembled carotenoids is normally related to hydrophobic interactions in solvent–water mixtures because water molecules do preferably built up extended intermolecular structures among themselves by hydrogen bonds and the carotenoid molecules are then forced to leave solvent spheres in which water is mainly present [12–14].

Since molecular aggregates have long been known from dye solutions, the basic properties of aggregates were derived from electronic absorption spectra by applying the pioneering approaches of Scheibe, Jelly and Davidov [15–17]. For spectroscopic investigations it is surely of particular interest to study aggregates in which molecules are highly ordered. Examples for well ordered systems are J-aggregates which are formed by head-to-tail positioned molecules and H-aggregates in which molecules are located face-to-face. An ideal head-to-tail arrangement in

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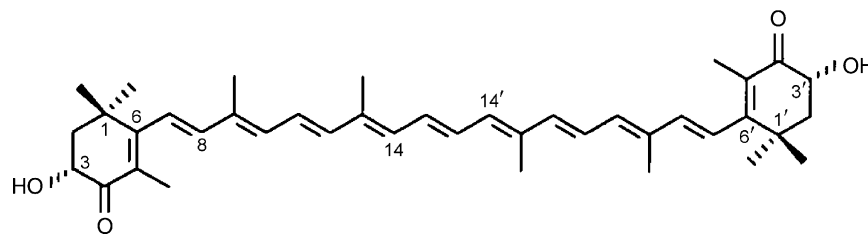


Fig. 1. The structure of *r,r'*-astaxanthin (3*R*,3'*R*-dihydroxy-4,4'-diketo- β -carotene).

J-aggregates leads to a parallel orientation of transition moments. The strong coupling among several monomers then results in a red-shifted transition relative to that of the monomer. In contrast to that, the face-to-face arrangement of H-aggregates causes a blue shift of the absorption band. While ideal J- and H-aggregates are one-dimensional arrangements and most suitable for theoretical considerations, the formation of aggregates more generally occur in ordered stacks along three axes. According to their spectral behavior these bulky assemblies are also denoted as J- and H-type aggregates.

This article is focused on the structure of H-aggregates of *r,r'*-astaxanthin in acetone–water mixtures. The assembled molecules were characterized by absorption spectra, CD (circular dichroism)-spectra and measuring fluorescence emission. The sizes of the molecular assemblies in solutions were determined by dynamic light scattering experiments. The experiments were complemented by molecular modeling calculations to derive the structure unit of the H-aggregated astaxanthin molecules. The calculated structure was confirmed by calculating the electronic absorption spectrum and the CD-spectrum where the molecular modeling structure was used as input. Finally, astaxanthin aggregates on graphite surfaces could be monitored by scanning tunneling microscopy.

2. Experimental section

Astaxanthin was kindly provided by BASF. The three stereo-isomers (3*S*, 3'*S*; 3*R*, 3'*S*; 3*R*, 3'*R*) were separated by using camphanoyl acid esters [18]. The samples were purified by column chromatography and the purification was checked by HPLC. The acetone–water mixtures (10^{-5} M) were prepared starting from a 10^{-4} M solution of the carotenoid dissolved in acetone.

The UV/vis absorption spectra were recorded using a HP 8452A diode array spectrometer (spectral resolution 2 nm; 1 cm block cuvette). The CD-spectra were obtained from a Jasco-600 spectropolarimeter (scan speed: 2 nm/min; path length: 2 cm).

The dynamic light scattering experiments were carried out by a light scattering goniometer (ALV, Langen, Germany) equipped with a correlator ALV 5000. This device was attached to a HeNe-Laser (633 nm, 40 mW). The equipment was checked by using a standard latex as

a sample. For evaluation of the time-correlation functions both the cumulative analysis and the CONTIN algorithm were applied [19]. The average diffusion coefficients obtained were transferred into an average hydrodynamic diameter $2R_h$ and a diameter distribution by formally using the Stokes–Einstein relationship for spheres.

The STM-images were achieved by a beetle-type microscope built by the Ingenieurbüro M. Wilms in cooperation with K. Wandelt (Institut für Physikalische Chemie und Theoretische Chemie, Universität Bonn). The tunneling current and the *z*-positioning of the tip were provided by galvanically separated 12-bit resolved ADCs. The accuracy of the bias setting was 0.1 mV. The *z*-position of the tip was calibrated by scanning steps on a HOPG surface. The constant current mode was used for scanning. Isolation from environmental vibrations was achieved by placing the microscope head on a granite slab which was held by springs. The tips were prepared by simply cutting a 0.2 mm Pt/Ir wire by a edge cutter. The HOPG samples (grade ZYH) were obtained from Burleigh Instruments and Kelpin Crystal Trading Inc., respectively. Flat and clean surfaces could easily be prepared by peeling upper layers with an adhesive tape. In order to achieve a homogeneous distribution of aggregates on HOPG surfaces, the STM sample holder was spinning during the covering of the substrate. The coating of the HOPG surfaces was obtained by evaporating acetone–water mixtures in which *r,r'*-astaxanthin was dissolved. The samples were dried under vacuum to eliminate remaining traces of the solvents.

Molecular modeling calculations were performed by using the program package Discover/Insight II (Molecular Simulation). The calculations of the aggregates based on the CVFF (consistent valence force field)-set-up. The optimization of structures starts with molecular dynamic simulations. In order to surmount conformational barriers within an initial geometry, the simulation temperature was chosen to 300 K while the time for the rearrangement of atoms was set to 500 ps. During the simulation, each 10 ps a conformational structure was stored. The 50 conformational structures were then virtually cooled down to 100 K in a period of 10 ps. This slow relaxation avoided the accidental freezing of unfavourable local minima within the potential surface. Finally, the 50 relaxed conformational structures were treated by the conjugate gradient routine with a termination criterion of 10^{-3} kcal Å⁻² in order to obtain the energy minimum of the chosen aggregate.

3. Results and discussion

Absorption spectra of carotenoids are generally shaped by a strong band in the visible region. This absorption band is attributed to the S_0 – S_2 transition [20,21]. The lower energetic S_0 – S_1 transition is not detectable by ordinary one-photon absorption spectroscopy. This common property of all-trans carotenoids is caused by the local C_{2h} symmetry of the polyene chromophore. The resulting selection rule implies that transitions between states of the same parity are dipole-forbidden. Since the first excited singlet state belongs to the same symmetry species A_g as the electronic ground state, the S_0 – S_1 transition is hidden for ordinary one-photon spectroscopy. The second excited singlet state transforms as B_u . That causes that the S_0 – S_2 transition is optically allowed.

The absorption maximum of the S_0 – S_2 transition of astaxanthin dissolved in acetone is recorded at 478 nm (Fig. 2). Adding water to the pure acetone solutions, the color of the sample changes significantly. Fig. 2 presents the spectra of two exemplary mixtures. Within the spectrum of the 1:9 acetone–water mixture, the maximum of absorption is recorded at 450 nm while the spectrum of the 3:7 acetone–water mixture exhibits two bands with absorption maxima at 520 and 562 nm. In mixtures with a high part of water, the S_0 – S_2 transition was blue-shifted. The opposite shift was observed in spectra of solutions with a large amount of acetone. Shifts of absorption bands could not be recorded from mixtures whose part of acetone was 50% or exceeded that value. The spectrum of astaxanthin in a 1:1 acetone–water mixture is nearly the same as recorded from a pure acetone solution.

In general, absorption spectra are changed by two solvent-induced effects. One, known as solvatochromy, is caused by the influence of the solvent as a dielectric medium and depends on the magnitude of the static dielectric constant and the refractive index of the solvent. S_0 – S_2

transition frequencies in spectra of carotenoids can be considerably affected by the choice of solvents [19]. However, varying the content of water in the mixtures, the changes of the refractive indices are too small to generate such a strong shifting of transitions as shown in Fig. 2. For this reason, the blue shift of the S_0 – S_2 transition is attributed to the formation of H-type aggregates and the two red-shifted transitions are assigned as absorption bands of two different J-type aggregates.

The presence of two different types of J-aggregates in 3:7 acetone–water mixtures can be proven by fluorescence measurements. Due to Davydov's excitonic model, J-aggregates are capable to fluoresce from the down-shifted state of lowest energy. Using 514.5 nm as excitation wavelength, the excitation energy exceeds the transition energies of both absorption bands. Considering, if both transitions belong to the spectrum of one species the resulting wavelengths of fluorescence have to be longer than 562 nm. Since the fluorescence emission begins at 543 nm as shown in Fig. 3, the absorption bands at 520 and 562 nm are attributed to two different J-aggregates. As it is common for S_2 – S_0 emissions of carotenoids, the fluorescence emission is weak [20,22]. The fluorescence emission of the J-aggregate is accompanied by ordinary Raman signals of the solvent molecules and resonance enhanced Raman signals of astaxanthin J-aggregate. Raman transitions of assembled carotenoid molecules can be well distinguished from those of monomer species by the shift of vibrational frequencies [23]. For instance, the signal of the central C=C stretching vibration of the polyene framework is recorded at 1516 cm^{-1} whereas the equivalent mode in the monomer spectrum appears at 1546 cm^{-1} .

Since astaxanthin contains OH groups in the terminating rings, it seems likely that the hydrogen bonds do also influence the formation of aggregates. This suggestion could

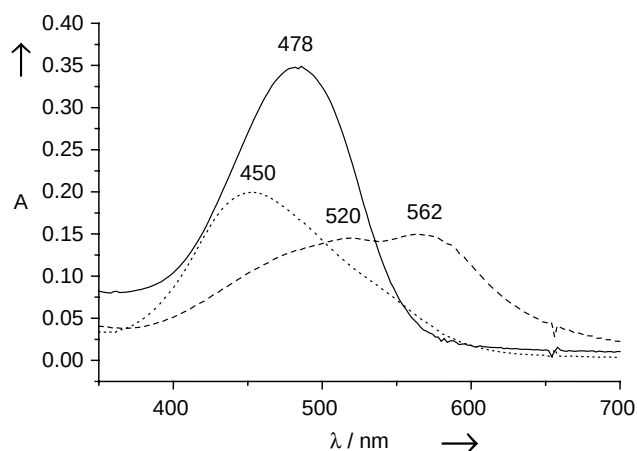


Fig. 2. UV/vis-spectra of *r,r*-astaxanthin in acetone/water mixtures. Solid line: acetone; 10^{-5} M; dotted line: 1:9 acetone–water mixture (10^{-5} M); dashed line: 3:7 acetone–water mixture (10^{-5} M).

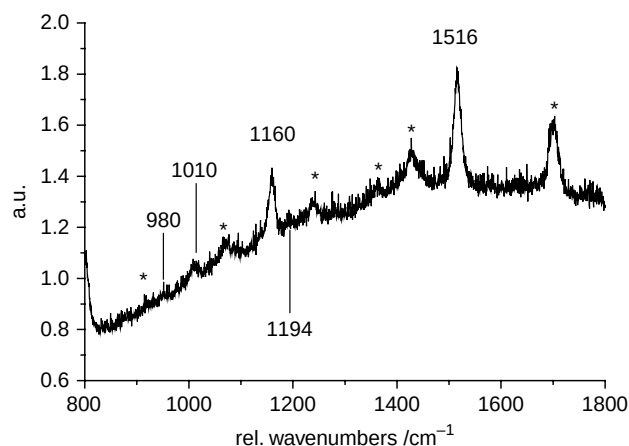


Fig. 3. Emission spectrum of *r,r*-astaxanthin. 3:7 acetone–water mixture (10^{-5} M); excitation wavelength: 514.5 nm ($19,436\text{ cm}^{-1}$); the abscissa denotes a scale of relative wavenumbers starting at 800 cm^{-1} (abs. $18,636\text{ cm}^{-1}$); Raman transitions of the J-aggregate are assigned by relative wavenumbers; Raman transitions of acetone are indicated by asterisks.

be confirmed by comparing the spectra of the diastereomers *r,r*-astaxanthin and *r,s*-astaxanthin (3*R*,3'*S*-dihydroxy-4,4'-diketo- β -carotene), each dissolved in acetone–water mixtures. While the absorption spectra of both diastereomers in pure acetonic solutions are undistinguishable, the spectrum obtained from *r,s*-astaxanthin in a 1:9 acetone–water mixture shows two different types of H-aggregates instead of only one as known from the corresponding spectrum of the *r,r*-astaxanthin. On the other hand, the spectrum of *r,s*-astaxanthin in a 3:7 acetone–water mixture cannot be distinguished from that of *r,r*-astaxanthin dissolved in the same mixture. The spectral feature originates apparently from two J-type *r,s*-astaxanthin aggregates which exhibit the same absorption spectrum as detected from the *r,r*-astaxanthin solutions.

The interpretation of CD spectra gives further information about the structure of assembled astaxanthin molecules. In principle, optical activity of monomeric carotenoids can be related to three sources (Fig. 1): the asymmetric C(3) atom, the conformation of the terminating rings and the angle between the polyene chain and the ring. The latter contribution is caused by a steric hindrance along the formal single bond C(6)–C(7). The twist along this bond originates a handedness because one conformational form of the ring is predominant. On the other hand, due to the local symmetry C_{2h} , the S_0 – S_2 transition is magnetic-dipole forbidden [24]. Moreover, the double bonds of the terminating rings are only partially conjugated with the polyene chain and the OH groups do not contribute to the chain chromophore. This means that the above mentioned sources of the optical activity are rather ineffective. The CD-spectrum of *r,r*-astaxanthin dissolved in pure acetone exhibits a small and nearly constant negative molar ellipticity. The optical activity under the intense S_0 – S_2 transition is for instance about -1 mdegree at 478 nm.

In contrast to the small molar ellipticity of astaxanthin when dissolved in pure acetone, acetone–water mixtures show a distinct circular dichroism. With regard to the aforementioned reasons for the missing optical activity, the recorded circular dichroism is therefore attributed to the overall structure of the assembled carotenoids. The CD-spectrum of astaxanthin in the 1:9 acetone–water mixture exhibits two Cotton effects (Fig. 4). The CD-spectrum reverts its sign at 436 nm near the maximum of the H-aggregate absorption at 450 nm. A positive circular dichroism is present for wavelengths smaller than 436 nm and a negative circular dichroism is recorded between 650 and 436 nm. Since for paired molecules the wavelength at which the reversal of sign occurs coincides with the maximum of absorbance, the aggregate studied is composed by more than two molecules. The different absolute values of the maximum and minimum ellipticity within the CD-spectrum show further that the orientation of monomer molecules within the H-aggregate is not strictly parallel. Additionally, due to the model of coupled oscillators introduced first by Kuhn [25], the observed

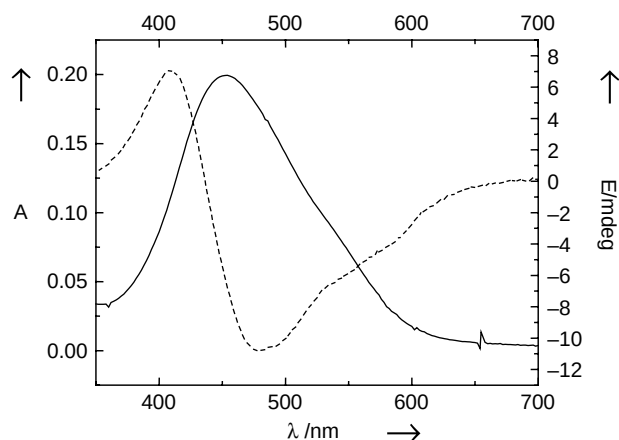


Fig. 4. CD-spectrum of the H-aggregate of *r,r*-astaxanthin (1:9 acetone–water mixture, 10^{-5} M) in relation to the absorption spectrum. dashed line: CD-spectrum (ellipticity E versus wavelength); solid line: absorption spectrum.

sequence of circular dichroism and its reversal of the sign refer to a left-handed helical structure of the H-aggregates.

Formations of supramolecular carotenoid assemblies are not limited to left-handed structures only. Both left-handed and right-handed aggregates have been found. For example 6'-epimers of capsanthol in ethanol–water mixtures do form right-handed as well as left-handed aggregates and right-handed helical structures of assembled carotenoids have been reported for lutein dispersed in bile–acid solutions [26,27].

The CD-spectrum of the 3:7 acetone–water mixture in which two different J-type aggregates are present, shows that each of the corresponding partial spectra exhibits two Cotton effects with differing signs (Fig. 5). The circular dichroism of the J-aggregate, whose recorded electronic transition has higher energy, has the same reversal of the sign as found for the H-aggregate. Therefore, a left-handed helical structure can be attributed to that J-aggregate.

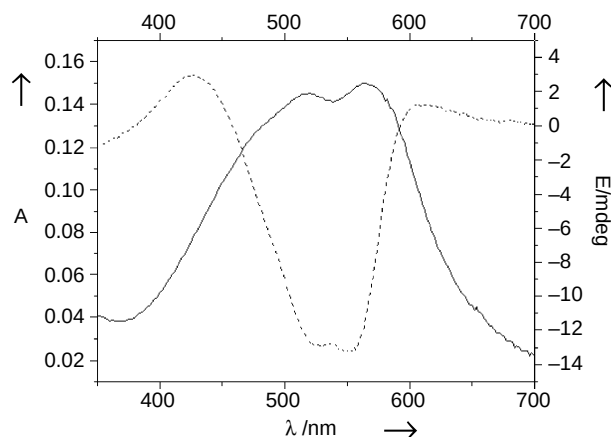


Fig. 5. CD-spectrum of the J-aggregates of *r,r*-astaxanthin. (3:7 acetone–water mixture, 10^{-5} M) in relation to the absorption spectrum. dashed line: CD-spectrum (ellipticity E versus wavelength); solid line: absorption spectrum.

On the other hand, the change of the ellipticity which is detected in the range of the lower energetic absorption band is opposite to that which is recorded for the two other astaxanthin assemblies. This indicates that both J-aggregates apparently have helices with opposite sense of rotation.

The structural results obtained by spectroscopic measurements for the astaxanthin H-aggregates were confirmed by molecular modeling calculations. The H-aggregate structure of *r,r*-astaxanthin was modeled by 12 monomers. The four central molecules are depicted in Fig. 6. Due to the card-packed formation, the stacking of the molecules occurs

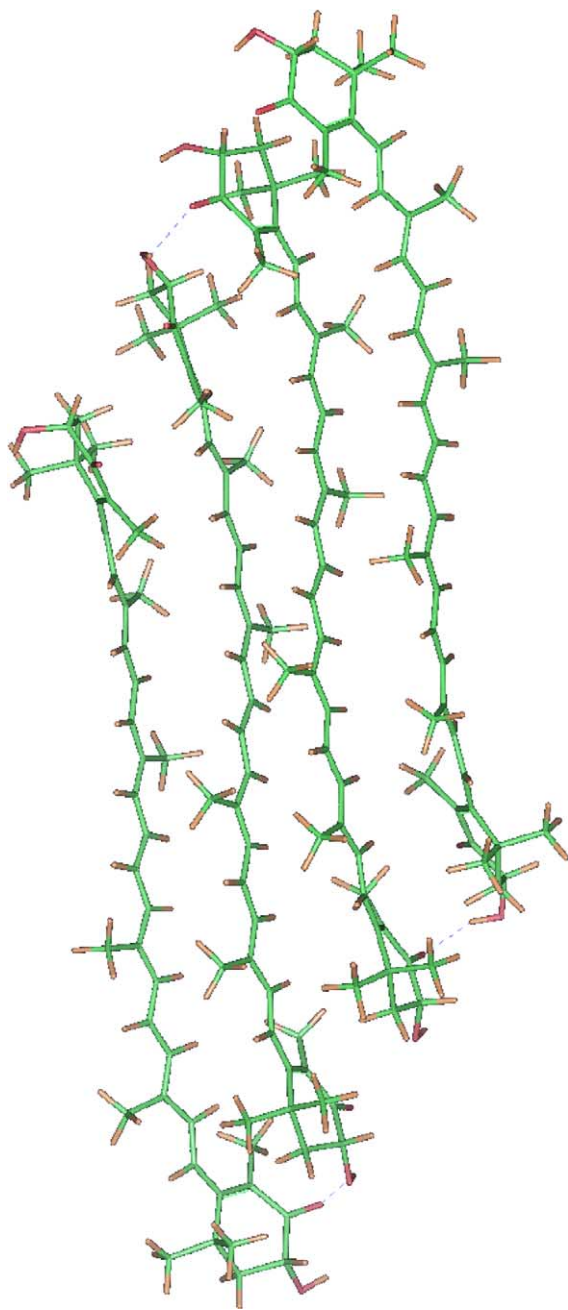


Fig. 6. Central part of the H-aggregate of *r,r*-astaxanthin obtained by molecular modeling. The broken lines indicate H-bonding.

perpendicular to the polyene planes. The nearest averaged interatomic C···C distance between to adjacent chains amounts to 420 pm. Next-neighbored molecules are not completely parallel. For example the two centered carotenoids are turned along the central axis at which stacking occurs. The rotational angle can be roughly estimated to 5°. Additionally, neighbored molecules are sheared along the chain directions. Using the methyl-group substituted atoms in the chain at the positions 9, 13 as references for one astaxanthin and 13 for the neighbored molecule, too, the angle of shearing amounts 40°. Hydrogen bonds occur between keto groups and OH groups. Each carotenoid has a slight s-shaped chain. The s-shape is marked by an angle of about 12° between the atoms 6, 11 and 15. The deviation from the linear polyene framework can be attributed to the repulsive forces of the methyl groups substituted along the chain. This property was first experimentally established by an X-ray analysis of β -carotene [28]. In total these geometric properties result in a left-handed helical structure of the assembly.

The structure of the H-aggregate obtained by the molecular modeling calculation was used as input geometry to calculate transition energies, oscillator strengths and the molar ellipticity using the program of V. Buß [29]. The program, which based on the CNDO/S-CI set-up, calculated the absorption spectrum and CD-spectrum of astaxanthin aggregate units. For computational restrictions the size of the aggregate was limited to 12 molecules. The calculations considered 60 microstates (a microstate defines an electronic configuration which is described by a single Slater determinant). The calculated absorption spectrum and CD-spectrum are presented in Fig. 7. The transition energies, the oscillator strengths and the magnitudes of circular dichroism converged already to constant values when the aggregates were composed by eight molecules.

Comparing the spectrum shown in Fig. 4 with the graphs in Fig. 7, theoretically derived spectra match the experimental spectra and it can therefore be concluded that

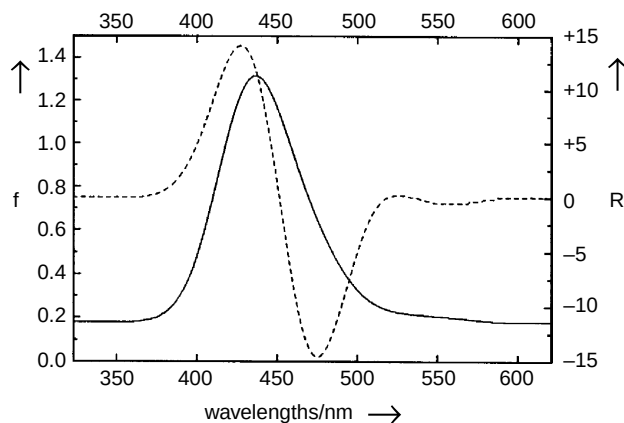


Fig. 7. The calculated CD- and absorption spectrum of the astaxanthin H-aggregate in relation to the spectra shown in Fig. 4. The computed absorptivity f and the rotational strength R are presented in arbitrary units.

the structure of the H-aggregate is quite well described by the structure obtained from the molecular modeling treatment.

Dynamic light scattering was applied to determine the sizes of the aggregates. The hydrodynamic diameter $2R_h$ of the H-aggregates, which originates from the 1:9 acetone–water mixtures, was determined to 40 nm. In mixtures with a higher amount of acetone the size of the aggregates becomes larger. The average $2R_h$ increases to 300 nm in 1:3 acetone–water mixture and exceeds up to 1 μm for particles in 3:7 acetone–water mixtures.

Mixtures with a content of acetone larger than 50%, did not contain measurable particles of assembled molecules. This means, with regard to the formerly presented absorption spectra, the recorded absorption bands can be attributed to monomeric species because aggregates with negligible excitonic splitting as possible for assemblies with magic shearing angle do not contribute to the shape of the absorption band.

Finally, astaxanthin assemblies could be directly monitored on HOPG (highly oriented pyrolytic graphite) surfaces by applying scanning tunneling microscopy. Fig. 8 presents a global view of a typical covering of the surface with assembled astaxanthin molecules obtained from a 1.9×10^{-5} M acetone–water mixture. Different rod-like structures could be detected. The sizes of the rods vary from 3 to 5 μm . Comparing the STM-images obtained from various 1:9 mixtures, the lengths and the widths of the rods alter randomly with each preparation of the coating.

Zooming the STM-images by reducing the scanning area, structural details in the rods become visible (Fig. 9). Each rod consists of parallel rows in which rod-shaped subunits are successively stringed. The extension of the subunits between the centers of the bright spots is 2.37 nm. This length corresponds well to the interatomic distance of 2.33 nm between C(5) and C(5') within a *r,r*-astaxanthin molecule according to molecular modeling calculations and semi-empirical treatments. The bright spots therefore roughly mark the position of the rings substituted to

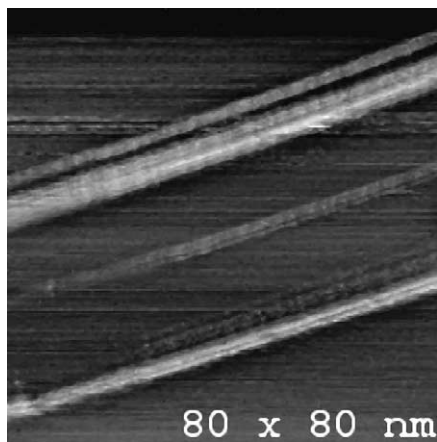


Fig. 8. STM-image of *r,r*-astaxanthin on HOPG. Scanning area as indicated at the right bottom of the image; bias: 800 mV; tunneling current: 2 nA.

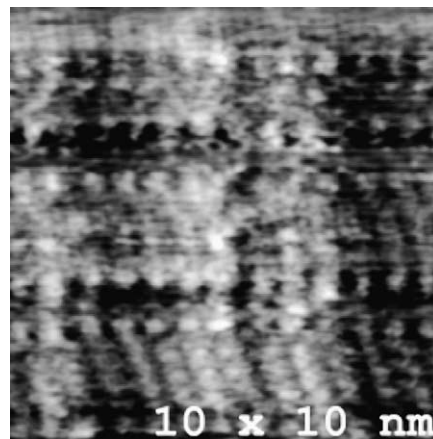


Fig. 9. Zoomed STM-image. scanning area as indicated at the right bottom of the image; bias: 800 mV; tunneling current: 2 nA.

the chain and this illustrates that the rings are located above each polyene chain. In this top view presentation, most of the astaxanthin molecules have bent chains but in some cases, a weak s-shaped structure is visible. These different appearances of the astaxanthin molecules may result from different mutual orientations of the molecules with respect to the plane of projection.

As expected from spectra of the 1:9 acetone–water solutions, the astaxanthin molecules are primarily arranged in a card-packed formation. The average distance between molecules within the card-packed arrangement amounts to about 0.59 nm, which is measured between the brightly appearing centers of adjacent polyene chains. The distance between neighbored rows is somewhat more than 1 nm. Since this spacing is too large for fixing molecules by intermolecular forces, it seems reasonable that at least one layer is beneath the top coating. A multilayer arrangement is first indicated by the arches within the top layer near the center of the depicted STM-image (Fig. 9). This is additionally revealed by plotting the tunneling current versus the distance across to the parallel chains. That plot shows two steps at the edge of the aggregate structure which means that at least three layers cover the substrate. Thus, the large gap between the rows of molecules may be explained by a shearing of carotenoids in adjacent layers causing a brick-work-like composition.

The distinct monitoring of the *r,r*-astaxanthin molecules indicates that molecular chains are deposited plane on the surface in contrast to formerly made STM-images of substituted polyenes. For instance, STM-images obtained from 7'-apo-7'-(4-mercaptomethylphenyl)- β -carotene which are embedded in monolayers of 1-docosanethiol and deposited on a gold surface show bright spots only [30]. This shape of STM-images was attributed to a structure in which the carotenoids are fixed to the gold atoms by their thiol groups and therefore the molecules stick out from the surface plane. A precondition for imaging these thiols is a high electric conductance along the chain of the carotenoids.

For the STM-images presented in Figs. 8 and 9 a good transverse conductance for astaxanthin molecules has also to be considered since more than one layer covers the HOPG surface.

Finally, reflecting the investigated astaxanthin assemblies, it seems most likely that astaxanthin molecules require an extraneous assistance for the formation of large aggregates for instance by the choice of solvents. For the chosen experimental conditions, the keto and hydroxyl groups which are normally predestined for forming intermolecular bonds seem to play a minor role within the process of self-assembling. A pure self-induced organization of astaxanthin molecules to aggregates should be less essential since both the CD-spectroscopy and the measurements of dynamic light scattering did not give any indications for the formation of aggregates in pure acetone solutions. This lack of building self-organized assemblies distinguishes carotenoids for instance from porphyrin molecules to which they have a close relation by their symbiosis within light harvesting complexes [31].

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